FINITE MORPHISMS ARE LIMITS OF FINITE FP MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2969-2981, following the statement at 2960-2967.

Editorial note. The final paragraph is a construction-reading step ('By construction ...') anchored to relative Spec and the algebra engine; flagged in the notes.

Allowed standard prerequisites (editorial).

- Finite algebras as directed colimits. lemma-finite-algebra-directed-colimit-finite-finitely-presented (same chapter, cited at ~2971-2973). A = f_*O_X = colim A_i with A_i finite fp O_Y-algebras, surjective transitions.
- Relative Spec. Morphisms of Spaces, Definition spaces-morphisms-definition-relative-spec (cross-chapter, cited at ~2974-2976). X_i = relative Spec_Y(A_i).

Source reference. Original lemma 086Z: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2639–2639. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081W: focused-seeds/geometry/S9-spaces-limits-finite-in-finite-and-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a finite morphism of algebraic spaces over S. Assume Y quasi-compact and quasi-separated. Then X can be written as a directed limit $X = \lim X_i$ where the transition maps are closed immersions and the objects X_i are finite and of finite presentation over Y.*

Proof. Consider the finite quasi-coherent $\mathcal{O}_Y$-module $\mathcal{A} = f_*\mathcal{O}_X$. By Lemma 086Z we can write $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ as a directed colimit of finite and finitely presented $\mathcal{O}_Y$-algebras $\mathcal{A}_i$ with surjective transition maps. Set $X_i = \underline{\operatorname{Spec}}_Y(\mathcal{A}_i)$, see Morphisms of Spaces, Definition 081W. By construction $X_i \to Y$ is finite and of finite presentation, the transition maps are closed immersions, and $X = \lim X_i$. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.